Econ 602 Notes

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Simultaneous-move Games of Complete Information

Can model these as strategic-form games

Def "strategic-form game"

A strategic-form game is a $G = (S_i, u_i = 1, ..., I)$ such that

1. $\{1, ..., I\}$ is a set of players
2. S_i is a set of strategies for player i
3. $u_i : \Pi_{j=1}^{I} S_j \to \mathbb{R}$ is a payoff function for player i. (u_i maps from strategy profiles to real (integrable) numbers.)

Example 1: introduce NE

Def "Nash Equilibrium (PSNE)"

Call $(s_1^*, ..., s_I^*)$ a Nash equilibrium if, for each $i = 1, ..., I$

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}), \forall s_i \in S_i.$$

"deviating to some s_i is not going to make me any better off.

Def "Best Response"

Say $s_i \in S_i$ is a best response for i under s_{-i} if

$$u_i(s_i, s_{-i}) \geq u_i(s_i', s_{-i}) \quad \forall s_i' \in S_i.$$

Def "Best Response Correspondence"

$BR_i : S_{-i} \rightrightarrows S_i$ such that

$$BR_i(s_{-i}) = \{s_i \in S_i : s_i \text{ is a best response for } i \text{ under } s_{-i}\}$$

Remark 1 A strategy profile is a Nash Equilibrium if and only if, for each $i = 1, ..., I$

$$s_i^* \in BR_i(s_{-i}).$$

Example 2 Maximize revenue to find BR correspondence. Determine PSNE from correspondence.

Example 3 "Increasing payoffs can hurt"

Difference between Game Theory and Decision Theory: In DT, increasing payoffs cannot hurt. In GT, there are scenarios under which increasing payoffs can hurt.

Example 4 "incentive to hide what you are playing"

Randomization is a way of hiding what's being played.

Def "set of mixed strategies of i"

The set of probability measures or distributions on S_i is $\Delta(S_i)$.

Given $E_i \subseteq S_i$ (event), $\sigma(E_i)$ is the probability that i plays a strategy in E_i, if i chooses the mixed strategy σ_i.

Remark 2 "infinite strategies have no atoms..." If S_i is finite, then $\sigma_i(E_i) = \sum_{s_i \in E_i} \sigma_i(s_i)$. But, this property need not hold when S_i is infinite.

Remark 3 We have (above) defined a new game called the "mixed extension" of $G = (S_i, u_i : i = 1, ..., I)$. The differences are (1) strategy sets are now $\Delta(S_i)$, and (2) the payoff of i from $(\sigma_1, ..., \sigma_I)$ is $u_i(\sigma_1, ..., \sigma_I)$.

Remark 4 Consider a finite game (a game in which S_i is finite for each $i = 1, ..., I$.). The payoff is

$$u_i(\sigma_1, ..., \sigma_I) = \Sigma_{S_I} \cdots \Sigma_{S_1} u_i(s_1, ..., s_I)\sigma_1(s_1) \cdots \sigma_I(s_I).$$

Def "MSNE"

Call $(\sigma_1, ..., \sigma_I)$ a "Mixed Strategy Nash Equilibrium" (MSNE) if, and only if, for each $i = 1, ..., I$

$$u_i(\sigma_i^*, \sigma_{-i}^*) \geq u_i(\sigma_i, \sigma_{-i}), \forall \sigma_i \in \Delta(S_i).$$

Lemma 1 "MSNE $\iff$ weakly better than all pure strategies"

A strategy profile $(\sigma_1, ..., \sigma_I)$ is a MSNE if, and only if, for each i

$$u_i(\sigma_i^*, \sigma_{-i}^*) \geq u_i(s_i, \sigma_{-i}^*), \forall s_i \in S_i.$$

Lemma 2 "MSNE $\iff$ strategy in support is best"

Fix a finite game $G = (S_i, u_i : i = 1, ..., I)$.

Then $(\sigma_1, ..., \sigma_I)$ is a MSNE if, and only if, for each i and for each $s_i \in S_i$, with $\sigma_i^*(s_i) > 0$,

$$u_i(s_i, \sigma_{-i}^*) \geq u_i(r_i, \sigma_{-i}^*), \forall r_i \in S_i.$$

Remark 5 "Lemma 2 doesn't have the right context for the infinite case"

Cor 1 "indifference condition"

Fix a finite game $G = (S_i, u_i : i = 1, ..., I)$.

If $(\sigma_1, ..., \sigma_I)$ is a MSNE and with $\sigma_i^*(s_i), \sigma_i^*(r_i) > 0$, then

$$u_i(s_i, \sigma_{-i}^*) = u_i(r_i, \sigma_{-i}^*).$$

Example 5 "Lemma 2 doesn't work in the infinite case + intuition for how to extend"

Thm "from online notes"

1. If σ_i induces F_i, then F_i is a CDF
2. If F_i is a CDF, then there exists a *unique* σ_i that induces F_i.

Remark 6 "unanswered questions" We constructed an example of a symmetric MSNE. But, we have not shown it is the only such symmetric MSNE. Can we have a symmetric MSNE with gaps in the support? Can we have a symmetric MSNE with atoms?

Remark 7 "pdf/cdf technical notes" omitted

Existence of Pure and Mixed Strategy Nash Equilibria

Notation: "Mixed Best Response Correspondence" We've already seen one correspondence: $BR_i : S_{-i} \rightrightarrows S_i$ such that

$$BR_i(s_{-i}) = \{s_i \in S_i : s_i \text{ is a best response to } s_{-i}\}.$$

Now, we have $\tilde{BR}_i : \Pi_{j \neq i} \Delta(S_j) \rightrightarrows \Delta(S_i)$ such that

$$\tilde{BR}_i(\sigma_{-i}) = \{\sigma_i \in \Delta(S_i) : \sigma_i \text{ is a best response under } \sigma_{-i}\}.$$

For all players:

$$\tilde{BR} : \Pi_{i=1}^I \Delta(S_i) \rightrightarrows \Pi_{i=1}^I \Delta(S_i) \text{ such that } \tilde{BR}_i(\sigma_1, ..., \sigma_I) = \Pi_{i=1}^I \tilde{BR}(\sigma_1, ..., \sigma_{i-1}, \sigma_{i+1}, ..., \sigma_I).$$

Result "NE $\iff$ fixed point of BR"

- $(s_1^*, ..., s_I^*)$ is a PSNE $\iff$ $(s_1^*, ..., s_I^*) \in BR(s_1^*, ..., s_I^*)$ $\iff$ fixed point of BR.
- $(\sigma_1^*, ..., \sigma_I^*)$ is a MSNE $\iff$ $(s_1^*, ..., s_I^*) \in \tilde{BR}(\sigma_1^*, ..., \sigma_I^*)$ $\iff$ fixed point of $\tilde{BR}$.

Thm "Kakutani's fixed point thm"

Let X be nonempty compact convex set. If F is nonempty, convex-valued, and has a closed graph, then F has a fixed point.

Thm 1 "conditions to guarantee existence of PSNE"

Fix a game where, for each $i = 1, ..., I$, if

1. there exists an n_i such that S_i is a nonempty, compact, convex subset of $\mathbb{R}^{n_i}$,
2. $u_i : S \to \mathbb{R}$ is continuous, and
3. for each s_{-i}, holding s_i fixed, $u_i(\cdot, s_{-i}) : S_i \to \mathbb{R}$ is quasiconcave.

   nonempty: from S_i being compact and continuity of utility fn (BR has to be nonempty)

   closed-graph: from continuity of u_i

   convex-valued (F): from quasiconcavity of $u_i(\cdot, s_{-i}) : S_i \to \mathbb{R}$

Thm 2 "conditions to guarantee existence of MSNE"

For each finite game, there exists a MSNE.

Thm 3 "Glicksberg's thm"

Fix a game, G where, for each i,

1. S_i is compact metric space, and
2. $u_i : S \to \mathbb{R}$ is continuous.

Then G has a MSNE.

Remark 8 Glicksberg's thm doesn't tell us that the assumptions of compactness and continuity are interesting. E.g., first-price auctions are discontinuous.

Remark 9 Glicksberg's thm only tells us there is some equilibrium–there may be many.

Remark 10 Existence results are most useful if there are qualitative results of how equlibria vary with interesting parameters of the model/game.

Rationalizability and Iterated Dominance

Example 1 "motivation for beliefs: can have wrong beliefs now."

Term "belief" Belief of player i: $\mu_i \in \Delta(S_{-i})$

Def "best response with beliefs"

Fix a nonempty set $Q_i \subseteq S_i$. Say s_i *is a best response under* μ_i *given* Q_i if $s_i \in Q_i$ and

$$\int_{S_{-i}} u_i(s_i, s_{-i}) d\mu_i \geq \int_{S_{-i}} u_i(r_i, s_{-i}) d\mu_i, \ \forall r_i \in Q_i.$$

Say s_i *is a best response under* μ_i if it is a best response under μ_i given S_i.

Or, in discrete settings:

$$\sum_{s_{-i} \in S_{-i}} u_i(s_i, s_{-i}) \mu_i(s_{-i}) \geq \sum_{s_{-i} \in S_{-i}} u_i(r_i, s_{-i}) \mu_i(s_{-i}), \ \forall r_i \in Q_i.$$

Def "justifiable"

Fix a nonempty (candidate subset) $Q_{-i} \subseteq S_{-i}$. Say s_i *is justifiable given* Q_{-i} if there exists $\mu_i \in \Delta(S_{-i})$ such that

1. s_i is a best response under μ_i and
2. $\mu_i(Q_{-i}) = 1$.

Say s_i is *justifiable* if it is justifiable given S_{-i}.

Def "rationalizable sets"

Let $R_i^0 = S_i; R_{-i}^0 = S_{-i}$.

Inductively define sets R_i^m and R_{-i}^m such that

- $R_i^{m+1} = \{s_i \in S_i : s_i \text{ is justifiable given } R_{-i}^m\}$, and
- $R_{-i}^{m+1} = \Pi_{j \neq i} R_j^{m+1}$.

Additional notation:

- $R_i^\infty = \cap_{m \geq 0}$ is the set of rationalizable strategies for i.
- $R^\infty = \Pi_{i=1}^I R_i^\infty$ is the set of rationalizable strategy profiles.
- $R^m = \Pi_{i=1}^I R_i^m$ is the set of $m-$rationalizable strategy profiles.

Examples 2-4 "which strategies are justifiable?"

Def "(un)dominated"

Fix $Q_1, ..., Q_I$ where each $Q_j \subseteq S_j$.

Say s_i *is dominated given* $\Pi_{j=1}^I Q_j$ if $s_i \in Q_i$ and there exists a $\sigma_i \in \Delta(S_i)$ such that

$$u_i(\sigma_i, s_{-i}) > u_i(s_i, s_{-i}),\ \forall s_{-i} \in \Pi_{j \neq i}^I Q_j.$$

Say s_i *is undominated given* $\Pi_{j=1}^I Q_j$ if it is not dominated given $\Pi_{j=1}^I Q_j$.

Say s_i *is (un)dominated* if it is (un)dominated given S.

Def "undominated sets"

Let $U_i^0 = S_i$.

Inductively define U_i^m such that

$$U_i^{m+1} = \{s_i \in S_i : s_i \text{ is undominated given } U_{-i}^m\}.$$

Additional notation:

- $U_i^\infty = \cap_{m \geq 0} U_i^m$ is the set of undominated strategies for i.
- $U^m = \Pi_{i=1}^I R_i^m$ is the set of $m-$undominated strategy profiles.
- $U^\infty = \Pi_{i=1}^I U_i^\infty$ is the set of *iteratively undominated* (IU) strategy profiles.

Thm 1 "$R^m = U^m$"

Fix a finite game. $R^m = U^m,\ \forall m$.

Prop 1 "justifiable $\iff$ undominated"

Fix a finite game. A strategy s_i is justifiable if, and only if, it is undominated.

Def "Hyperplane"

A hyperplane in $\mathbb{R}^n$ is a set of the form

$$H[\vec{p}, \alpha] = \{\vec{x} \in \mathbb{R}^n : \vec{p} \cdot \vec{x} = \alpha\}$$

for $\vec{p} \in \mathbb{R}^n \setminus \{0\}$ and $\alpha \in \mathbb{R}$.

Def "supporting hyperplane"

Fix $A \subseteq \mathbb{R}^n$ and $\vec{x} \in A$. Say $\vec{p} \in \mathbb{R}^n$ (with $\vec{p} \neq \vec{0}$) *supports A at $\vec{x}$* if

$$\vec{p} \cdot \vec{x} \geq \vec{p} \cdot \vec{y}, \forall y \in A.$$

Say $H[\vec{p}, \alpha]$ supports A at $\vec{x}$ if $\vec{p}$ supports A at $\vec{x}$ and $\vec{p} \cdot \vec{x} = \alpha$.

Thm "Supporting Hyperplane Theorem"

Let $A \subseteq \mathbb{R}^n$, convex and $\vec{x} \in A$. There exists some $\vec{p} \in \mathbb{R}^n$ (with $\vec{p} \neq \vec{0}$) such that $\vec{p}$ supports A at $\vec{x}$ if and only if $\vec{x} \notin int(A)$. (i.e., $\vec{x} \in$ boundary(A))

Example 6 "weakly dominated"

Lemma 1 "later sets subsets of earlier subsets"

For each $i = 1, ..., I$ and each $m \geq 0$, $R_i^{m+1} \subseteq R_i^m$.

Cor 0.1 "rationalizable set stops shrinking at some point"

If the game is finite, there exists some number M such that $R^m = R^M$ for all $m \geq M$.

Example 0.1 NEED TO WORK THROUGH

Prop 1 "nonempty rationalizable set exists if compact and continuous"

Fix a game $G = (S_i, u_i : i = 1, ..., I)$ where each S_i is a compact metrics space and each u_i is continuous. Then the set of rationalizable strategy profiles is nonempty.

Def "Weakly Dominated"

Fix non-empty sets $Q_1, ..., Q_I$. Say s_i is *weakly dominated given* $\Pi_{j=1}^I Q_j$ if $s_i \in Q_i$ and there exists $\sigma_i \in \Delta(Q_i)$ so that

1. $u_i(\sigma_i, s_{-i}) \geq u_i(s_i, s_{-i})$ for each $s_{-i} \in \Pi_{j \neq i} Q_j$, and
2. $u_i(\sigma_i, s_{-i}) > u_i(s_i, s_{-i})$ for some $s_{-i} \in \Pi_{j \neq i} Q_j$.

Say s_i is *weakly dominated* if it is weakly dominated given S.

Def "Iterated Weak Dominance"

Set $W_i^0 = S_i$. Inductively define W_i^m such that

$$W_i^{m+1} = \{s_i \in W_i^m : s_i \text{ is not weakly dominated given } \Pi_{j=1}^I W_j^m\}.$$

Then W_i^{m+1} is the set of strategies of i that survive $(m+1)$ rounds of iterated weak dominance.

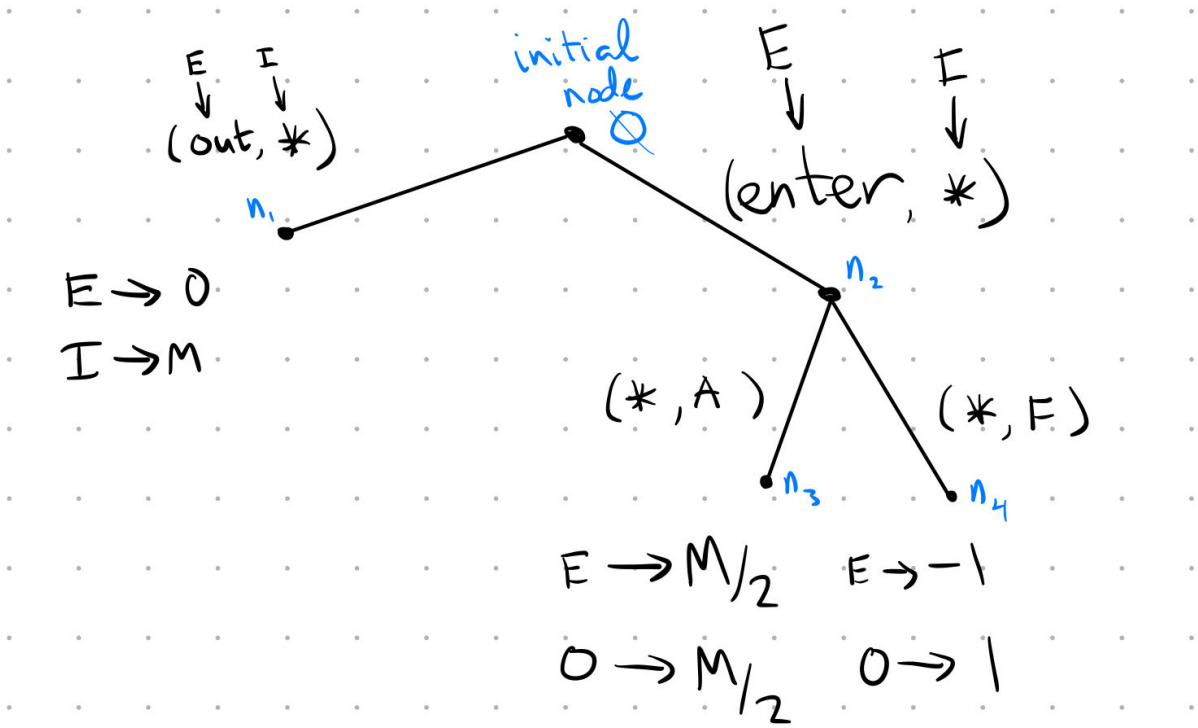
Set $W_i^\infty = \cap_{m \geq 0} W_i^m$. Then $W^\infty = \Pi_{i=1}^I W_i^\infty$ is the set of strategy profiles that survive iterated weak dominance.

Sequential Games of Complete Information

Example 1 Entrant (E) can *enter* or stay *out*. If E enters, then incumbent monopolist (I) can **F**ight or **A**cquiesce.

Profits to (E, I), subject to $M \geq 2$, are:

- (out, $*$): $(0, M)$
- (enter, A): $(\frac{M}{2}, \frac{M}{2})$
- (enter, F): $(-1, 1)$



Remark 1 Some people will see an example like example 1 and conclude "the strategic form game does not suffice." This is incorrect. We will see (later) that the problem is the solution concept of Nash Equilibrium, not the strategic form, per se.

Visualization vs Formalization - omitted for now.

Def Extensive Form (model a sequential game as an extensive-form game)

We model a sequential game as an extensive form game: $\Gamma = (A1, ..., A_J; N; H_1, ..., H_I; U_1, ..., U_I)$.

Extensive-Form 1: Players $1, ..., I$

Extensive-Form 2: Action Sets $(A_1, ..., A_I)$, where A_i is the set of actions player i can choose *somewhere* in the game. E.g., $A_E = \{O, E, *\}, A_I = \{F, A, *\}$

Extensive-Form 3: Nodes N = set of nodes.

> Each non-terminal node, n is associated with a set $A_i(n)$ =set of actions of i that can be played at n.
>
> $\Pi_{i=1}^{I} A_i(n)$ =set of action profiles that can be played at n.
>
> $A_i(n) \neq \phi$ but might be a singleton (e.g., $A_i(\phi) = \{*\}$).
>
> \$Z = \$ set of terminal nodes. We don't define $A_i(n)$ at terminal nodes.

Think of nodes as being defined by sequences of action profiles. E.g.,

- initial node = the empty sequence, ϕ, because nothing has been played.
- $n_1 = (O, *)$
- $n_2 = (E, *)$
- $n_3 = ((E, *), (*, A))$

- $n_4 = ((E, *), (*, F))$

Extensive-Form 4: Information Sets $H_1, ..., H_I$

- H_i is the set of information sets for i
  - if $h_i \in H_i$ then $h_i \subseteq N \setminus Z$
- H_i forms a partition of $N \setminus Z$
- If $n, n' \in h_i, h_i \in H_i$ then $A_i(n) = A_i(n')$

Extensive-Form 5: Payoff Functions

Mapped from terminal nodes to numbers:

$$U_i : Z \to \mathbb{R}$$

Typically add

1. No absentmindedness (rules out forgetting if you moved at all), and
2. Perfect recall (rules out a situation where I remember I moved, but not what I chose).

Term "No absentmindedness" Intuitively, this requires that i does not forget that she made *some* previous choice. So, there is no $h_i \in H_i$ with $n, n' \in h_i$ so that n strictly precedes n'. (If there were such an n and n', then at n' player i has forgotten that she was at n and made a choice.)

Term "Perfect recall" Intuitively, this requires that i remembers *which* previous choice they made. So there is no situation where, say, i knows they moved previously but cannot remember if they chose l or r.

Def "Specific Games of Interest"

1. Simultaneous-move games: the only non-terminal node is ϕ.
2. Games with observable actions: for each player, i, each $h_i \in H_i$ is a singleton.
3. Perfect-info games: games with observable actions where, at each info set there is at most one active player

Term "Active player"

$$|A_i(n)| \geq 2$$

Example 2 "Introduces (derived) strategies"

Def "strategy"

A strategy for i is an $s_i : H_i \to A_i$ such that $s_i(h_i) \in A_i(h_i)$.

(Note: we've only allowed pure strategies so far, but mixed will fit in later).

A strategy is a full contingency plan of action. It tells me what to do at each info set.

A strategy will specify 1 action at EACH $h_i \in H_i$ where i is making a choice.

Term "Path function"

$$\zeta : S \to Z$$

Term "strategic-form payoff function"

$$u_i : S \to \mathbb{R} \text{ such that } u_i(s) = U_i(\zeta(s)), \forall s \in S.$$

Example 3: In a simultaneous game: $S_i = A_i$.

Analyzing games: Solution concepts

- we can always write down the strategic form of the game and solve by NE. The problem is that NE doesn't require sequentially optimal behavior.
- instead, we need sequential optimality and for the players to believe other players are also sequentially optimizing.

Def "subgame"

We identify subgames with singleton information sets, $h = \{n\}$ where

1. h is an information set for all i, and
2. if n' follows n, and $n' \in h'_i$, then all nodes in h'_i follow n.

Facts about subgames:

- Each subgame is associated with a new strategic-form game which is associated with h. We call this new, strategic form game G^h (Remember, G for strategic/normal, and Γ for sequential/extensive).
- Intuitively, G^h consists of strategies in the *original* strategic-form game that allow h.
- Given $s_i \in S_i$, write s_i^h for a (potentially different) strategy that allows h and otherwise agrees with s_i.

Def "Subgame Perfect Equilibrium (SPE)"

Call $(s_1^*, ..., s_I^*)$ a *subgame perfect equilibrium* (SPE) if, for each subgame h, $(s_1^{*h}, ..., s_I^{*h})$ is a (pure strategy) Nash equilibrium of G^h.

Example 3 Battle of the sexes with an outside option.

Object "Backward-induction algorithm"

1. Look at each last nonterminal node and the player who moves there. Find *an* action that is a best response for that player at that node.
2. Look at nonterminal nodes that are followed only by "last nonterminal nodes". Find an action (for the player who moves at such a node) that is a best response given the behavior specified in step 1.
3. Repeat until we reach the start of the game.

This delivers an action associated with each information set. Collect these actions into strategies of each player and we have an SPE.

(re-run the algorithm for each action that is a best response at each node to find the full set of SPE)

Caution: Backward-induction applies only to perfect information games.

Example 4 "SPE without observable actions may look peculiar"

Important point: for the purpose of iterated weak dominance, it suffices to look at the "reduced strategic form"

Term "reduced strategic form" The strategic form that collapses all realization-equivalent strategies into one "strategy"

Def "behavioral strategy"

A *behavioral strategy* of i is a $b_i : H_i \to \Delta(A_i)$ such that

$$\forall h_i \in H_i, b_i(h_i)(A_i(h_i)) = 1.$$

E.g., $b_2(h)(l) = P(l|h) = \frac{1}{4} + \frac{1}{2}$ and $b_2(h')(l) = P(l|h') = \frac{1}{4} + \frac{1}{8}$.

Example 5 Extending SPE to behavioral strategies (note: we do this in example only because the notation is cumbersome.)

Misc

Def "pareto dominate"

Say a strategy profile $(s'_1, ..., s'_I)$ Pareto dominates $(s_1, ..., s_I)$ if

1. for all i, $u_i(s'_1, ..., s'_I) \geq u_i(s_1, ..., s_I)$ and
2. for some i, $u_i(s'_1, ..., s'_I) > u_i(s_1, ..., s_I)$.